

HWS in R Studio

Cole Chisamore

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1 Introduction

In this pdf, I will using R to attempt to model the response variable housing price based on multiple explanatory variables:

- Bedrooms: Number of bedrooms in the property (main or otherwise)
- Bathrooms: Number of bathrooms in the property (main or otherwise)
- Square Feet: Rough square footage of indoor property
- Property Type: Category assigned to each listing including...
 - Apt/Condo
 - Duplex
 - Fourplex
 - House
 - Land/Lot
 - Mfd/Mobile Home
 - Multifamily
 - Recreational
 - Townhouse
 - Triplex

2 Forming the Initial Model

I will begin with a naive model after reading in the scraped data. However I will first remove duplicates and invalid responses, and factorize Property Type. The model will be a basic generalized linear model correlating Price with each explanatory variable.

```
library(tidyverse)
library(forcats)
library(car)
library(rstatix)
## Polishing and Sampling dataframe:
houses.df <- read_csv(file = "housing_df.csv", show_col_types = FALSE)
houses.df <- houses.df %>% drop_na() %>% distinct()
houses.df$`Property Type` <- factor(houses.df$`Property Type`)
houses.df <- houses.df %>% mutate(`Property Type` = fct_reorder(
  `Property Type`, Price, .fun="length", .desc = TRUE))
houses.df <- houses.df %>% rename(PropertyType = `Property Type`)
houses.df <- houses.df %>% rename(SquareFeet = `Square Feet`)
## Creating first model:
houses.glm1 <- glm(formula = Price~Bedrooms+Bathrooms+SquareFeet+PropertyType,
  data=houses.df)
#' Prints a concise summary of model
#'
#' @param model A linear model
#' @param name String
my_summary <- function(model, name="Unnamed Model"){
  cat("\n",name,"\n\n")
  cat("Formula: ",as.character(model$formula)[c(2,1,3)],"\n\n")
  printCoefmat(summary(model)$coefficients)
  cat(paste("\nAIC:", model$aic))}
my_summary(houses.glm1, "Naive Model")
```

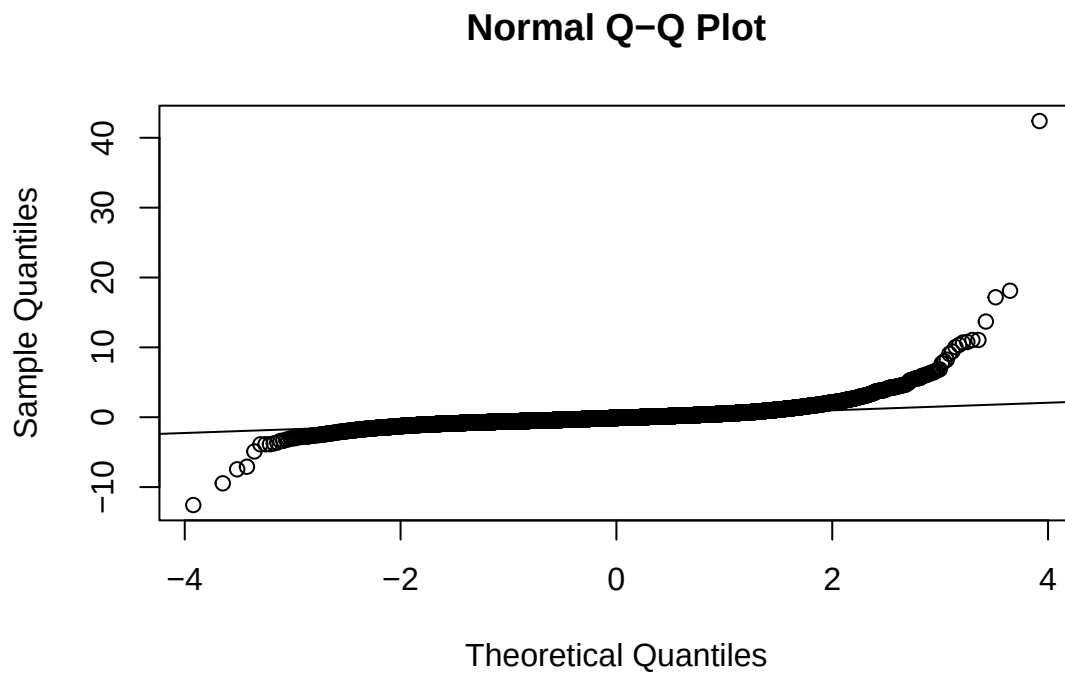
```
## " Naive Model "
## Formula: Price ~ Bedrooms + Bathrooms + SquareFeet + PropertyType
##
##
## Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.5304e+05 1.8902e+04 -8.0966 6.230e-16 ***
## Bedrooms -5.8731e+04 5.3270e+03 -11.0251 < 2.2e-16 ***
## Bathrooms 7.1028e+04 7.2835e+03 9.7519 < 2.2e-16 ***
## SquareFeet 5.9355e+02 9.7947e+00 60.5987 < 2.2e-16 ***
## PropertyTypeApt/Condo 1.9560e+05 1.6159e+04 12.1044 < 2.2e-16 ***
## PropertyTypeTownhouse -2.0045e+04 1.7440e+04 -1.1493 0.2504431
## PropertyTypeDuplex 1.2096e+05 3.0172e+04 4.0090 6.138e-05 ***
## PropertyTypeMfd/Mobile Home -2.2625e+05 6.1389e+04 -3.6855 0.0002293 ***
## PropertyTypeMultifamily -9.7978e+04 7.4030e+04 -1.3235 0.1856981
## PropertyTypeFourplex -3.2229e+05 1.8282e+05 -1.7629 0.0779487 .
## PropertyTypeLand/Lot -1.4119e+05 1.9142e+05 -0.7376 0.4607782
## PropertyTypeTriplex -1.0399e+05 2.0192e+05 -0.5150 0.6065605
## PropertyTypeRecreational -3.0259e+05 3.4919e+05 -0.8665 0.3862114
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## AIC: 333764.193110199
```

Before analyzing, let's also check the error of the model and create a qqplot of residuals for trained model:

```
## Calculates percent error between predictions and actual values
##
## @param preds predicted values
## @param reals actual values
error <- function(preds, reals){
  return(mean(abs(preds - reals) / reals))
}
pred <- predict(houses.glm1, houses.df)
error(pred, houses.df$Price)
```

```
## [1] 0.526457
```

```
houses.glm1.res <- rstandard(houses.glm1)
qqnorm(houses.glm1.res)
qqline(houses.glm1.res)
```



2.1 Initial Model Remarks

- A percent error of ~53% is not great, so hopefully I can create a better model later on
- The Q-Q plot shows heavy tails off of the normal line, especially on the right
 - This implies that the 'Price' variable is not normally distributed enough with many outliers (especially high outliers)
 - Perhaps removing these outliers and/or transforming Price could help
- the Bedrooms variable is oddly assigned a negative (and significant) correlation with Price, which is counter-intuitive since more bedrooms would seem to imply high cost
- multiple Property Type categories are deemed to have insignificant difference from "House", perhaps I can reduce the number of categories or fix them some other way

3 Augmenting the Variables

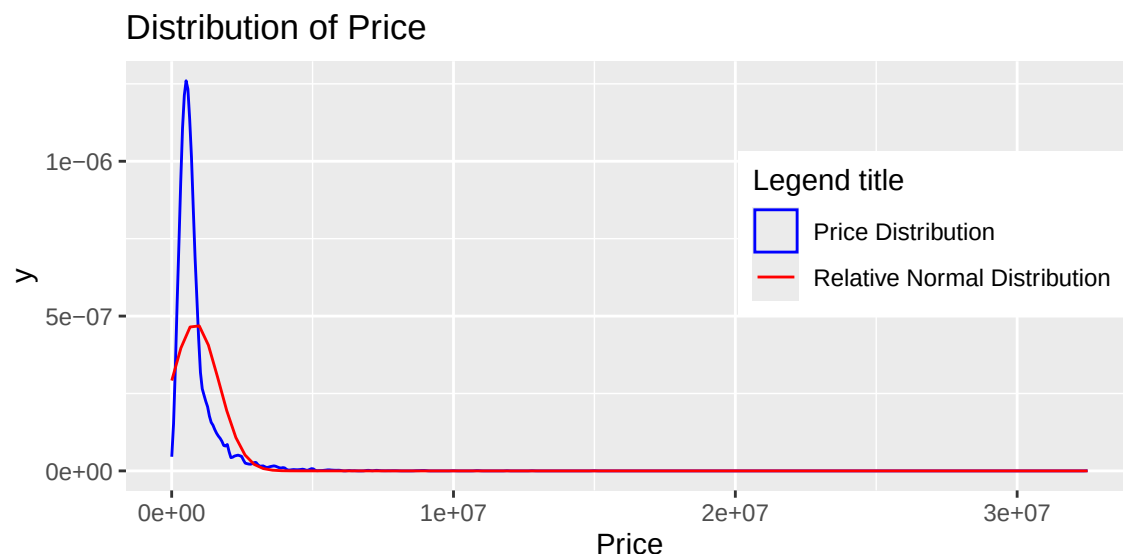
In this section I will begin augmenting the variables of my dataframe through transformation or dataframe manipulation in order to better fit the assumptions of the generalized linear model.

3.1 Price

3.1.1 Investigating the Variable

Since Price is the response variable of our model, it will be the most important one to have normally distributed for my glm. Let's begin by creating a density plot to see the current distribution:

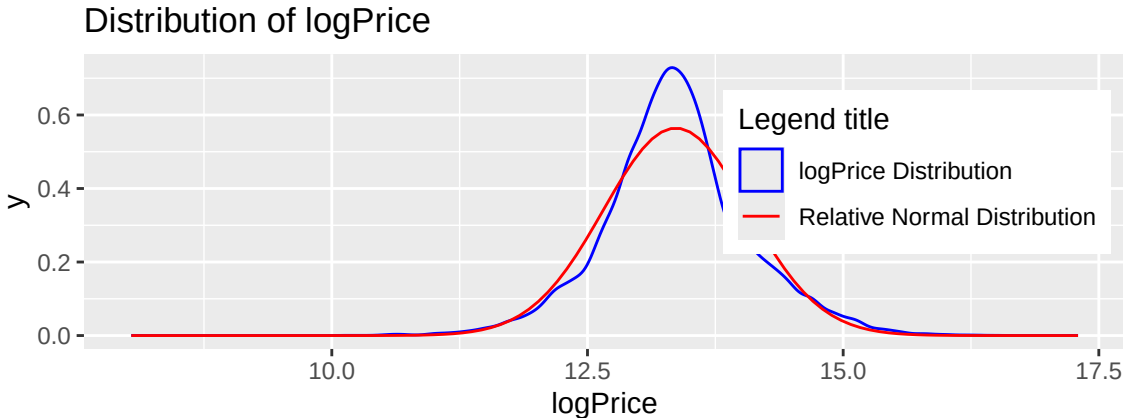
```
#' Creates a distribution plot object with relative normal distribution
#'
#' @param df A Dataframe
#' @param column A string representing a column name
#' @param datatype A string denoting plot type
#' @return distribution plot
distribution_graph <- function(df, column, datatype="bar") {
  y <- if(datatype=="density"){
    geom_density(aes(colour=paste(column,"Distribution")))
  } else {geom_bar(aes(colour=paste(column,"Distribution")))}
  p <- ggplot(data=df, aes(x=.data[[column]])) +
    ggtitle(paste("Distribution of", column)) + y +
    stat_function(fun=dnorm, aes(colour = "Relative Normal Distribution"),
                 args=list(mean=mean(df[[column]]), sd=sd(df[[column]]))) +
    scale_colour_manual("Legend title", values = c("blue", "red")) +
    theme(legend.position="inside", legend.position.inside = c(0.8, 0.6))
  return(p)
}
# Create the plot
distribution_graph(houses.df, "Price", datatype="density")
```



As was already seen in the naive model's Q-Q plot, the distribution of prices in our dataset is not very normal. There is a very heavy right-sided tail which makes this density curve much thinner than the relative normal curve that it should be. However, this should be easily fixed by a simple log transformation on Price:

3.1.2 Assessing Changes

```
## Create new "log(Price)" column for dataset and replot:
houses.df$logPrice <- log(houses.df$Price)
distribution_graph(houses.df, "logPrice", datatype="density")
```



This log transformation of Price seems to fit the normality assumption significantly better than before, so it will be used in place of Price. I will also remove all listings with extreme logPrice to cut down on tails.

3.1.3 Testing New Model

Let's recreate the model with my change and compare with the previous model.

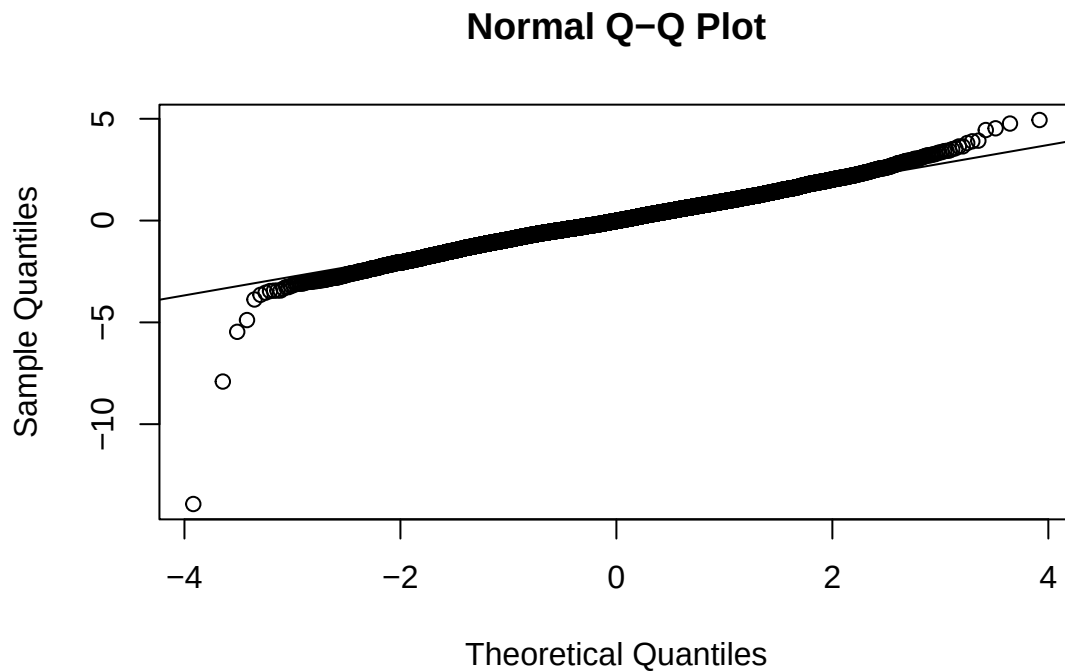
```
# Removing outlying logPrices, and making second model:
q99 <- quantile(houses.df$logPrice, probs=c(0.005,0.995))
houses.df <- houses.df %>% filter(logPrice > q99[1]) %>% filter(logPrice < q99[2])
houses.glm2 <- glm(formula = logPrice~Bedrooms+Bathrooms+SquareFeet+PropertyType,
                   data=houses.df)
my_summary(houses.glm2, "Model 2")
```

```
## " Model 2 "
## Formula: logPrice ~ Bedrooms + Bathrooms + SquareFeet + PropertyType
##
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.2379e+01 1.7774e-02 696.4828 < 2.2e-16 ***
## Bedrooms      1.1262e-02 5.3386e-03   2.1096  0.03491 *
## Bathrooms     1.2645e-01 6.6275e-03  19.0789 < 2.2e-16 ***
## SquareFeet    3.6484e-04 8.8103e-06  41.4105 < 2.2e-16 ***
## PropertyTypeApt/Condo 1.8674e-01 1.4307e-02  13.0522 < 2.2e-16 ***
## PropertyTypeTownhouse 3.5735e-02 1.4722e-02   2.4273  0.01523 *
## PropertyTypeDuplex   1.1306e-01 2.4881e-02   4.5440 5.578e-06 ***
## PropertyTypeMfd/Mobile Home -6.7277e-01 5.5150e-02 -12.1990 < 2.2e-16 ***
## PropertyTypeMultifamily -4.9819e-02 6.1014e-02  -0.8165  0.41422
## PropertyTypeFourplex  -2.0009e-01 1.5003e-01  -1.3337  0.18233
## PropertyTypeLand/Lot    1.1812e-01 1.6539e-01   0.7142  0.47512
## PropertyTypeTriplex     6.0312e-03 1.6560e-01   0.0364  0.97095
## PropertyTypeRecreational -2.4992e-01 2.8625e-01  -0.8731  0.38263
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## AIC: 16088.6192903172
```

```
pred <- exp(predict(houses.glm2, houses.df))
error(pred, houses.df$Price)
```

```
## [1] 0.4926671
```

```
houses.glm2.res <- rstandard(houses.glm2)
qqnorm(houses.glm2.res)
qqline(houses.glm2.res)
```



Remarks

- A percent error of ~49% is already a great improvement on the previous model
- The AIC score decreased drastically. However this is mostly due to the 'logPrice' variable being magnitudes smaller than the 'Price' variable, so no remarks can be made on AIC yet.
- The Q-Q plot still has noticeable tails curving away from the normal line
 - The right tail is much less heavy than before, an acceptable distance from the normal line
 - The left tail is about the same, with a few points straying quite further away
 - Overall the points are much straighter and close to the normal line
- Bedrooms now is being considered to have significant positive correlation, which is more reasonable
- Most Property Types still are considered insignificant

Overall, the model has improved in many ways and logPrice will be used in all future models.

3.2 Property Type

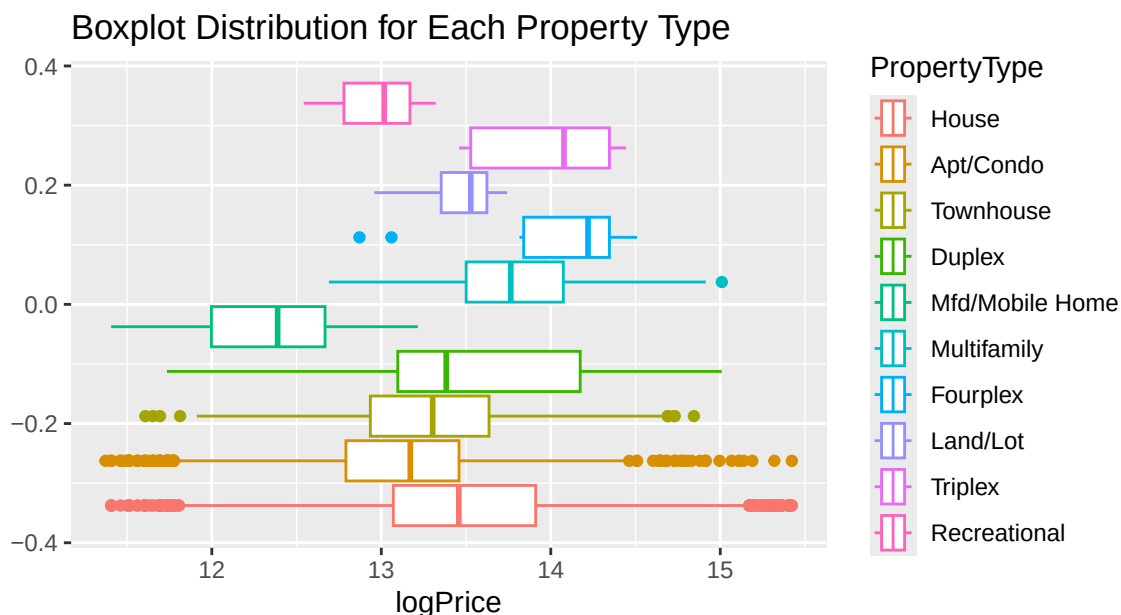
3.2.1 Investigating the Variable

Let's check the various property types and their distributions so we can see which ones are similar enough to be grouped, or scarce enough to be removed.

```
houses.df %>% group_by(PropertyType) %>% summarise(count=n())
```

```
## # A tibble: 10 x 2
##   PropertyType count
##   <fct>      <int>
## 1 House      5938
## 2 Apt/Condo  2966
## 3 Townhouse  1688
## 4 Duplex     436
## 5 Mfd/Mobile Home  83
## 6 Multifamily  69
## 7 Fourplex   11
## 8 Land/Lot    9
## 9 Triplex     9
## 10 Recreational 3
```

```
ggplot(data=houses.df, aes(x=logPrice, color=PropertyType)) +
  geom_boxplot() + ggtitle("Boxplot Distribution for Each Property Type")
```



Firstly the 'Land/Lot' and 'Recreational' categories are very scarce. Since 'Land/Lot' isn't really a kind of housing, I will simply remove any of these listings. 'Recreational' is also an odd type of housing and it has no similar categories, so it too will be removed.

Secondly since Triplex, Fourplex, and Multifamily are all on the higher end of the plot and are similar categories anyways, I will consider combining them (and perhaps other categories) after performing an ANOVA and a Tukey HCD test.

```
# removing 'Land/Lot' and 'Recreational' listings
houses.df <- houses.df %>%
  filter(!(PropertyType %in% c("Land/Lot", "Recreational")))
houses.df$PropertyType <- droplevels(houses.df$PropertyType)
```

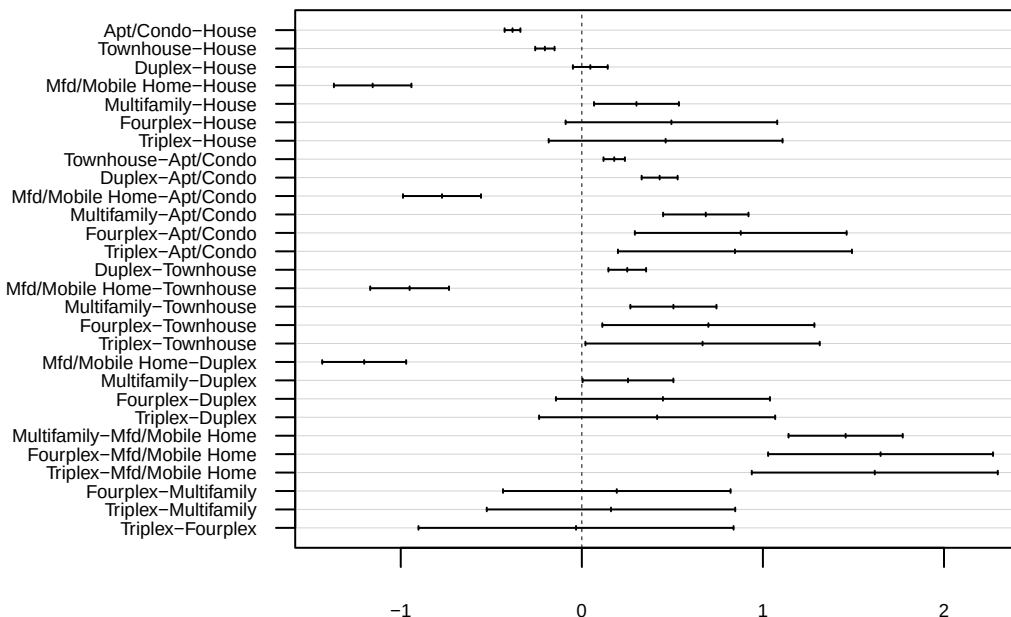
```
## ANOVA test
anova_model <- aov(logPrice ~ PropertyType, data = houses.df)
summary(anova_model)
```

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## PropertyType      7    420    59.99   147.2 <2e-16 ***
## Residuals     11192    4562     0.41
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Clearly there is significant difference between categories, but it was good to check anyway.

```
## Tukey HCD test
tukey_results <- TukeyHSD(anova_model, conf.level = 0.95)
par(mar=c(3,8,3,3))
plot(tukey_results, las=1, cex.axis = 0.6)
```

95% family-wise confidence level



Looking at which categories are/aren't significantly different in their mean, I can make some remarks:

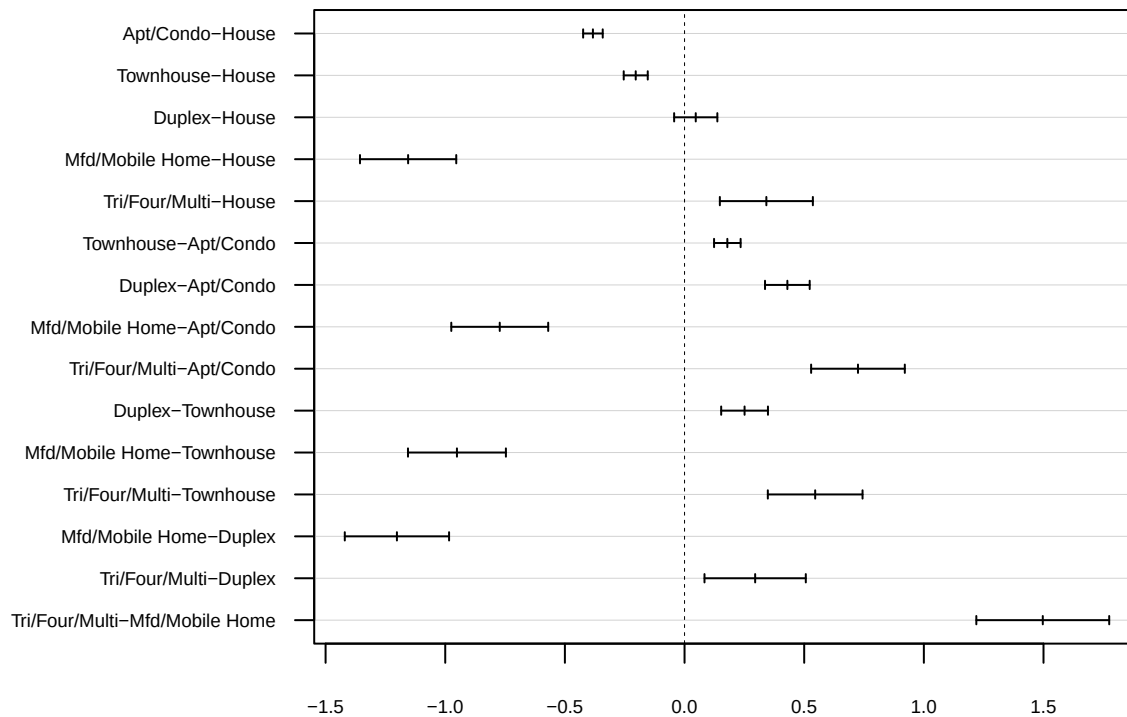
- Triplex, Fourplex, and Multifamily are all considered to not have significantly different means as expected.
- However Duplex is also considered to not have significantly different mean than these categories
- House is similar to Duplex in this regard, but IS significantly different from Multifamily
- House and Duplex both have similar distributions, and their means aren't significantly different

3.2.2 Making & Assessing Changes

Since Triplex, Fourplex, and Multifamily are so similar, I will try combining them into one category. Then I will re-plot and assess the changes.

```
# Merging categories
houses.df$PropertyType <-
  fct_collapse(houses.df$PropertyType,
    `Tri/Four/Multi` = c("Triplex", "Fourplex", "Multifamily"))
## Tukey HCD test
anova_model <- aov(logPrice ~ PropertyType, data = houses.df)
tukey_results <- TukeyHSD(anova_model, conf.level = 0.95)
par(mar=c(3,8,3,3))
plot(tukey_results, las=1, cex.axis = 0.6)
```

95% family-wise confidence level



It seems that the new category 'Tri/Four/Multi' has a mean significantly different from that of 'Duplex', so I will not be merging these categories. However the 'Duplex' and 'House' categories are still similar in mean (and distribution), so I will create two models: one with 'House' and 'Duplex' as separate categories, and one with them merged. Afterwards I will compare models and pick the one I find most suitable.

3.2.3 Testing New Models

Let's recreate the model with my change and compare with the previous model.

```
# Making model with House and Duplex separate
houses.glm3.1 <- glm(formula = logPrice~Bedrooms+Bathrooms+SquareFeet+PropertyType,
                     data=houses.df)
# Making model with House and Duplex merged
houses.df.2 <- houses.df
houses.df.2$PropertyType <-
  fct_collapse(houses.df.2$PropertyType,
               `House/Duplex` = c("House","Duplex"))
houses.glm3.2 <- glm(formula = logPrice~Bedrooms+Bathrooms+SquareFeet+PropertyType,
                     data=houses.df.2)
my_summary(houses.glm3.1, "Separate House/Duplex")
```

```
## " Separate House/Duplex "
## Formula: logPrice ~ Bedrooms + Bathrooms + SquareFeet + PropertyType
##
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.2378e+01 1.7788e-02 695.8480 < 2.2e-16 ***
## Bedrooms       1.1640e-02 5.3404e-03  2.1796  0.02931 *
## Bathrooms      1.2708e-01 6.6330e-03 19.1585 < 2.2e-16 ***
## SquareFeet     3.6405e-04 8.8122e-06 41.3118 < 2.2e-16 ***
## PropertyTypeApt/Condo 1.8760e-01 1.4311e-02 13.1090 < 2.2e-16 ***
## PropertyTypeTownhouse 3.5881e-02 1.4722e-02  2.4371  0.01482 *
## PropertyTypeDuplex   1.1273e-01 2.4882e-02  4.5306 5.943e-06 ***
## PropertyTypeMfd/Mobile Home -6.7205e-01 5.5151e-02 -12.1856 < 2.2e-16 ***
## PropertyTypeTri/Four/Multi -6.3848e-02 5.4045e-02 -1.1814  0.23748
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## AIC: 16067.5433356167
```

```
my_summary(houses.glm3.2, "Merged House/Duplex")
```

```
## " Merged House/Duplex "
## Formula: logPrice ~ Bedrooms + Bathrooms + SquareFeet + PropertyType
##
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.2386e+01 1.7697e-02 699.9307 < 2e-16 ***
## Bedrooms       1.0752e-02 5.3415e-03  2.0130  0.04414 *
## Bathrooms      1.3122e-01 6.5754e-03 19.9563 < 2e-16 ***
## SquareFeet     3.5886e-04 8.7450e-06 41.0356 < 2e-16 ***
## PropertyTypeApt/Condo 1.7847e-01 1.4180e-02 12.5853 < 2e-16 ***
## PropertyTypeTownhouse 2.5687e-02 1.4562e-02  1.7640  0.07777 .
## PropertyTypeMfd/Mobile Home -6.7970e-01 5.5173e-02 -12.3193 < 2e-16 ***
## PropertyTypeTri/Four/Multi -6.9914e-02 5.4076e-02 -1.2929  0.19608
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## AIC: 16086.0671962351
```

```
pred.1 <- exp(predict(houses.glm3.1, houses.df))
error(pred.1, houses.df$Price)
```

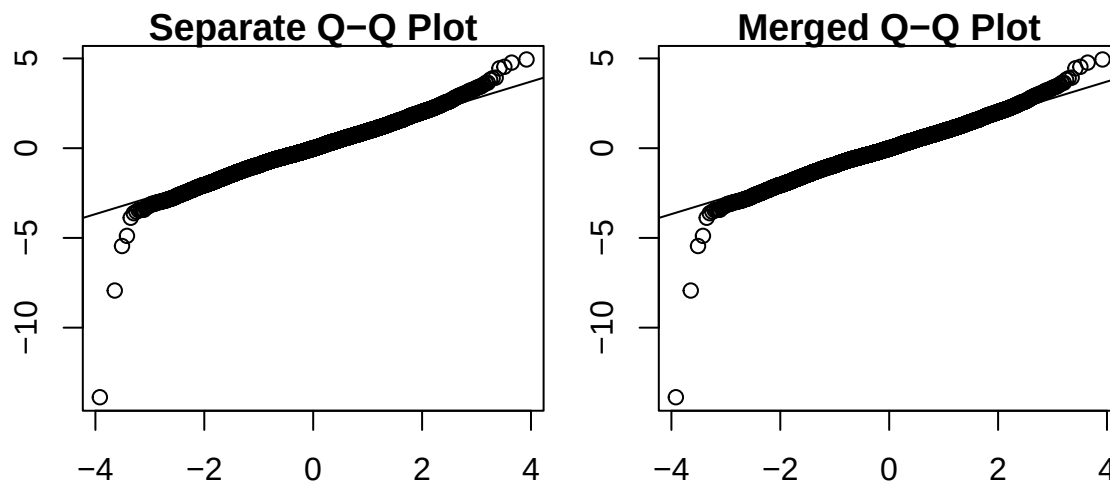
```
## [1] 0.4923533
```

```
pred.2 <- exp(predict(houses.glm3.2, houses.df.2))
error(pred.2, houses.df$Price)
```

```
## [1] 0.4928401
```

```
par(mar=c(2,2,1,1), mfrow = c(1, 2))
houses.glm3.1.res <- rstandard(houses.glm3.1)
qqnorm(houses.glm3.1.res, ylab = "", xlab = "", main="Separate Q-Q Plot")
qqline(houses.glm3.1.res)

houses.glm3.2.res <- rstandard(houses.glm3.1)
qqnorm(houses.glm3.2.res, ylab = "", xlab = "", main="Merged Q-Q Plot")
qqline(houses.glm3.2.res)
```



Remarks

- The AIC of both models is an improvement over the previous model
 - This implies that the removal and Tri/Four/Multi merging did in fact remove unnecessary variables
 - The AIC of the separate model is lower than the merged model, implying that merging is bad
- The Q-Q plots seem to be the exact same, and no different from the previous model's
 - This means the change didn't effect the normality of the errors
- The percent error of both models is very similar to the previous model (~49%)
 - The separate model actually slightly improves (by about 0.03%)
 - The merged model is slightly worse (by about 0.01%)

From these observations, it is clear that the separate model is better, and will become the newest model. I will note that there still exists a insignificancy problem with the 'Tri/Four/Multi' category, but I feel there are too many existing observations to simply remove this category from the dataset.

```
# Choosing official 3rd model
houses.glm3 <- houses.glm3.1
```

3.3 Numerical Explanatory Variables